# Deel Analytics Engineering Challenge

Edsel Volante
github.com/evolante03/deel-analytics-challenge

# Preliminary Data Exploration

## Acceptance report

- 5,430 rows, one per transaction
- external_ref is unique: the primary key
- 3,777 accepted / 1,653 declined
- 6 countries × 905 rows, Jan–Jun 2019

## Chargeback report

- 5,430 rows, joins 1:1 via external_ref key
- 223 rows with chargeback = TRUE (about 4%)
- status and source are constants in both files

## Notes

**Local currency:** Amounts must be converted to USD before aggregation

**One negative amount:** −23.78 GBP on an accepted transaction.

**API Spec difference:** The spec puts the evt_ id in external_ref; the data has it in ref. Also it was mentioned that external_ref is the card expiry year but it looks like a transaction reference id.

# Propose Model Architecture: sources → staging → mart

**SOURCES**
`globepay.raw`

- acceptance_report
- chargeback_report
- test for uniqueness and nulls

**STAGING · views**
`stg_globepay__*`

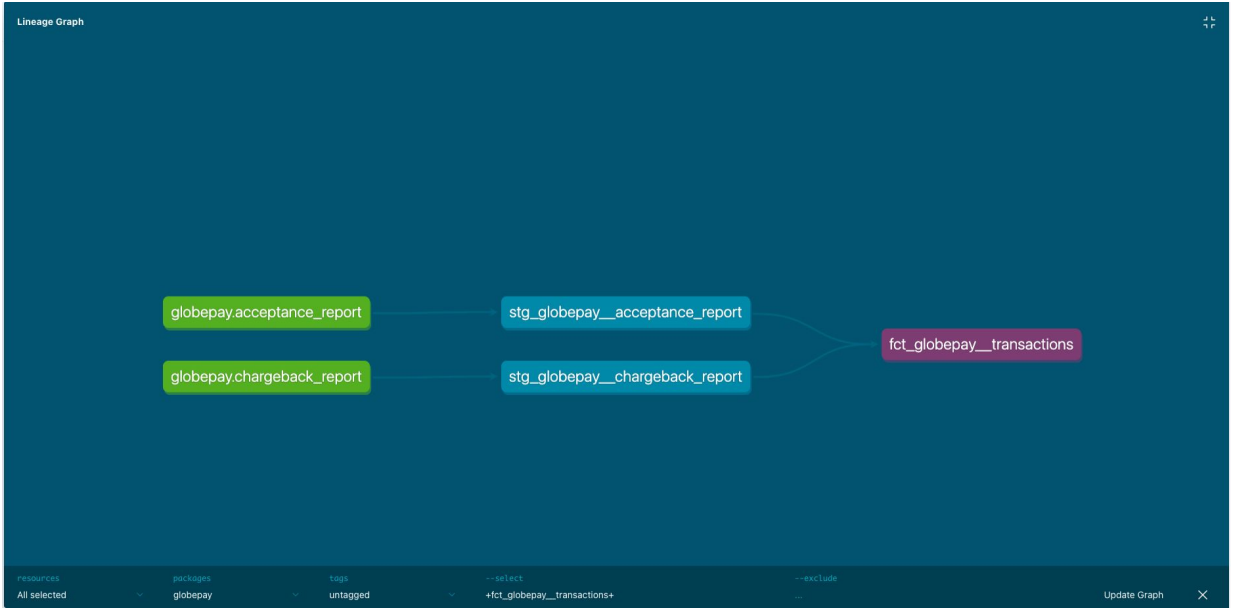
- Rename and cast columns
- Drop status and source columns
- Parse exchange rate rate from JSON
- amount_usd via convert_to_usd() macro

**MART · table**
`fct_globepay__transactions`

- One row per transaction
- Left join adds is_chargeback
- Null = missing chargeback record

*No intermediate layer since there are only two sources with 1:1 join.*

# Lineage from dbt docs



# Macros, validation and documentation

## Macros

convert_to_usd(amount, rate)

keeps the exchange rate logic in one place.

Other tips:
Keep macros small with simple inputs as much as possible

## Data validation

Different tests at source, staging and mart.

Errors for anything that breaks the numbers while warnings for things to investigate, like negative amounts or constants changing.

## Documentation

Model and column descriptions sit in schema.yml beside the models, so dbt docs shows them with lineage.

Naming conventions (stg_, fct_, is_, _ts, _usd) for ease of readability.

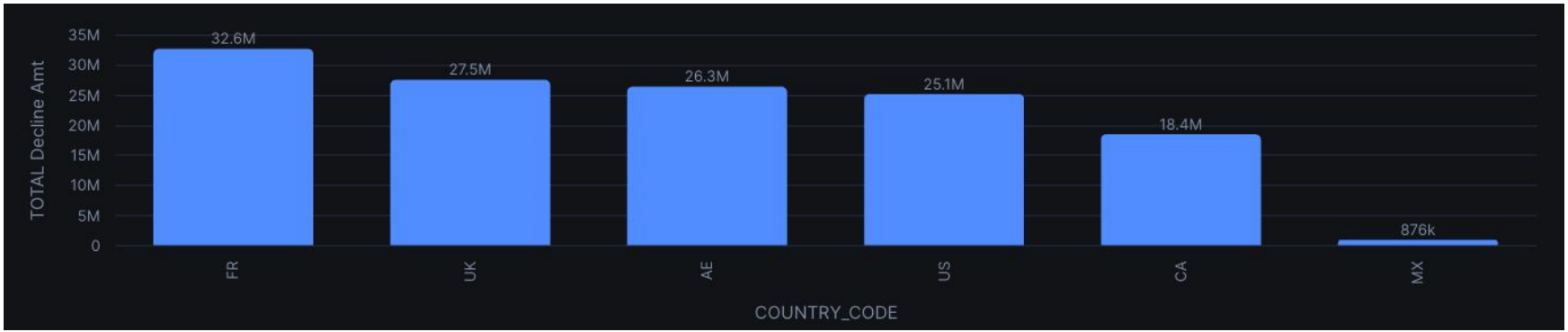
# Q1. Acceptance rate holds steady near 70%




Lowest in April (67.8%), highest in June (71.7%). Overall 69.6%.

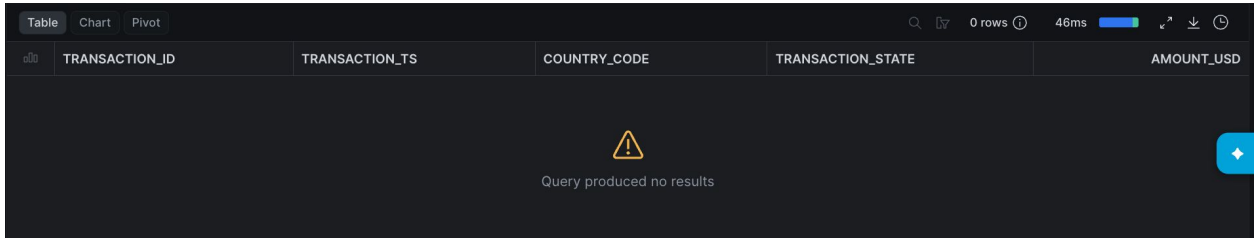
For other granularity we can swap 'month' for 'week' or 'day'  in the sql

# Q2. Four countries with total amount of declined transactions exceeded $25M




It is important that conversion is done first. When the local amount is aggregated in local currency, MX's total declined amount would look like $22M instead of $876k incorrectly.

# Q3. Every transaction has chargeback data




**Built to catch gaps.** The fact table left joins chargebacks onto transactions, so a transaction without a chargeback record is kept with is_chargeback = null rather than dropped.

**Null means missing.** The raw chargeback flag is tested as never null, so a null in the fact table can only mean the record is absent.

# Appendix: SQL

Q1

```
select
    date_trunc('month', transaction_date) as
transaction_month,
    count(transaction_id) as total_transactions,
    count(case when transaction_state =
'ACCEPTED' then transaction_id end) as
accepted_transactions,
    round(
        count(case when transaction_state =
'ACCEPTED' then transaction_id end)
        / count(transaction_id) * 100,
        1
    ) as acceptance_rate_pct
from
globepay.analytics.fct_globepay__transactions
group by 1
order by 1;
```

Q2

```
select
    country_code,
    count(transaction_id) as
declined_transactions,
    sum(amount_usd) as declined_amount_usd
from
globepay.analytics.fct_globepay__transactions
where transaction_state = 'DECLINED'
group by 1
having sum(amount_usd) > 25000000
order by declined_amount_usd desc;
```

Q3

```
select
    transaction_id,
    transaction_ts,
    country_code,
    transaction_state,
    amount_usd
from
globepay.analytics.fct_globepay__transactions
where is_chargeback is null
order by transaction_ts;
```